

# Modeling a Bayesian Network for a Diabetes Case Study

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**Abstract**—The prevalence of diabetes is increasing among all ages in the European Region. Treatment should be supported by a healthy diet, regular physical activity, maintaining healthy body weight, and avoiding tobacco use. If not managed correctly, sufferers are likely to become progressively more ill.

In this paper, a causal Bayesian network model is used for a diabetes case study analysis. The construction and evaluation of it have been done using Netica software. With the help of different probabilities allocated to the nodes, the results assigned to the graph reveals which nodes play a vital role in the onset of diabetes and even further in reaching other serious complications. Moreover, our study validates both the diabetes international criteria and the theory related to the transmission of information in cases of pieces of evidence.

**Keywords**—diabetes; Bayesian network; probabilities;

## I. INTRODUCTION

Diabetes, also known as diabetes mellitus, describes a group of metabolic disorders in which a person has blood glucose (blood sugar), either because insulin production is inadequate, or because the body's cells do not respond properly to insulin or even both. According to the World Health Organization, there are about 60 million people with diabetes in the European Region. Worldwide, high blood glucose kills about 3.4 million people annually [1-4].

Romania is among the first countries in Europe in terms of the incidence of this disease, over 2 million Romanians suffering from this disease [3-4].

About 90% of all cases of diabetes worldwide are of type 2 diabetes. The main risk factors for this type are unhealthy diet, physical inactivity, family history, age, high blood pressure, and abnormal cholesterol and triglyceride levels.

The symptoms of diabetes mellitus vary depending on the amount of sugar in the blood. Some of the common signs and symptoms of type 2 diabetes are permanent feelings of thirst, frequent urination, extreme hunger, fatigue, blurred vision, frequent skin, or urinary tract infections.

Patients are usually treated with medication, exercise, and a special diet, control of blood pressure, and cholesterol level, but sometimes insulin injections are needed. If diabetes is not adequately controlled, the patient has a significantly higher risk of developing complications, damaging the heart, blood vessels, kidneys, eyes, and nerves. Common consequences of uncontrolled diabetes include as well skin infections, neuropathy, ulcers, and sometimes gangrene.

The medical field is one of the areas in which the diagnostic process is particularly difficult to model and which raises many issues. The complexity and difficulty of modeling the medical decision-making process lie in the uncertainty that characterizes the medical knowledge regarding the possible diseases and the data regarding the patient's current condition, as well as the different approaches to the diagnosis, the doctor's reasoning in the diagnostic process.

This article proposes to model reasoning under uncertainty, by model a Bayesian network for a diabetes case study. A variety of influences of parent nodes values on the conditional probabilities of the node *Diabetes* is compared in order to see in what terms it reaches the highest probabilities values for this disease and other possible complications like cardiovascular disease, kidney failure, or diabetic neuropathy.

## II. BAYESIAN NETWORKS. RELEVANT WORK

In everyday life, in almost all areas of activity, we are put in the situation of making decisions in conditions of uncertainty. Uncertainty is one of the inherent and frequent properties of knowledge due to incomplete, inaccurate information or the use of approximate models to simplify problem-solving. Even if we reduce the uncertainty by gathering new information, we can rarely eliminate it.

Bayesian networks represent classes of probabilistic models used to model reasoning under conditions of uncertainty. They are directed acyclic graphs in which nodes represent random domain variables and directed arcs represent probabilistic dependencies between the nodes and their parents [5-6]. Within any Bayesian network, there are at least 3 nodes and at least one of the 3 types of connections: serial, convergent, divergent.

In a Bayesian network, the correct connection between the causal structure and the independent probabilistic is given by the Markov condition and the separation criterion. Once the wavelength probabilities of the variables are specified for all possible combinations of their parents, Bayesian networks have the property that uniquely defined a factorization of the composite probability distribution:

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1:n} P(X_i | X_{PA}^i) \quad (1)$$

where  $(X_1, X_2, \dots, X_n) = \text{domain variables}$  and  $X_{PA}^i = \text{parents nodes for each variable}$ .

Various implementations of Bayesian Networks ontologies have been made so far, not only for modeling diseases but for analyzing different genes expression to make connections between the most relevant genes that lead to the bad proliferation of cells in the human body.

In [7] a collaborative software agent framework with three key components is presented, having Bayesian Networks used as graphical models for probabilistic knowledge representation and uncertainty reasoning, while [8] provides an application created for medicine assisted diagnosis based on ontology and Bayesian Networks.

In [9] a Bayesian network model of heart disease is created. In [10], the authors proposed a Bayesian network decision model for supporting the diagnosis of dementia, using a combination of expert knowledge and data-oriented modeling. In [11], dynamic relations between the infectious diseases were identified based on dynamic Bayesian networks, while [12] provides a Bayesian network model for psychiatric disease diagnosis.

The authors in [13] applied a Bayesian belief network to estimate the risk of death in Hand, Foot, and Mouth disease based on pathogen detection and environmental factors.

### III. MODELING AND SIMULATION METHOD

Netica is a simulation environment for the construction and evaluation of Bayesian networks. It has the advantage of an intuitive user interface and offers computational support for achieving several types of inference based on modern and fast algorithms [7]. Netica offers flexibility, both in defining the relationships between variables and how to display the inference results.

Using Netica, we modeled a Bayesian network for diabetes mellitus, based on statistics available on the World Health Organization and American Diabetes Association (ADA) criteria.

The creation of our directed acyclic graph involves two main steps:

➔ Construction of the qualitative component, i.e. the directed acyclic graph, containing directed nodes and arcs (which implies the analysis of the problem and deduction of the causal relations between the variables);

➔ Defining the quantitative component, assuming the tables of conditional probabilities attached to each node of the network and describe the uncertainty on the (in)dependent relations between the respective node and its direct parents.

Our Bayesian network starts with three variables representing diabetes principal causes. ADA criteria for diabetes diagnosis uses mainly 2 indicators: basal blood glucose and body mass level intervals. Medically speaking if blood glycemic level  $\geq 126$  mg/dL and body mass index  $\geq 25$  Kg/m<sup>2</sup>, then that person is considered to be a diabetic. Being overweight and obese (BMI over 25 kg/m<sup>2</sup>) is strongly linked to the development of type 2 diabetes (for about 65–80% of new cases), but not everyone with type 2 is overweight. Also,

blood pressure above 140/90 millimeters of mercury (mm Hg) is linked to an increased risk of type 2 diabetes.

Table 1 shows the division of body mass index and glucose level.

TABLE I. BODY MASS INDEX AND GLUCOSE LEVEL

| IMC(KG/M <sup>2</sup> ) | VALUES     |
|-------------------------|------------|
| OPTIMUM WEIGHT          | 18.5-24.9  |
| OVERWEIGHT              | $\geq 25$  |
| GLUCOSE LEVEL           |            |
| NORMAL                  | 60-99      |
| ALTERED                 | 100-125    |
| HYPERGLYCEMIA           | $\geq 126$ |

The proposed Bayesian network finishes with some effects of this disease. According to the World Health Organization, the main consequences of type 2 diabetes are cardiovascular disease, kidney failure, and diabetic neuropathy.

In Figure 1 can be viewed the created graph. There are 9 serial connections, 3 convergent connections, and 3 divergent connections. Variables ‘body mass index’ (bmi), ‘blood pressure’ (bp), and ‘blood glycemic level’ (bgl) are parents for ‘diabetes’ (d) and ‘diabetes’ is the parent for the last three nodes (‘cardiovascular disease’ (cd), ‘kidney failure’ (kf), ‘diabetic neuropathy’ (dn)) of our Bayesian network. there can be seen as well, the marginal probabilities attributed to all nodes.

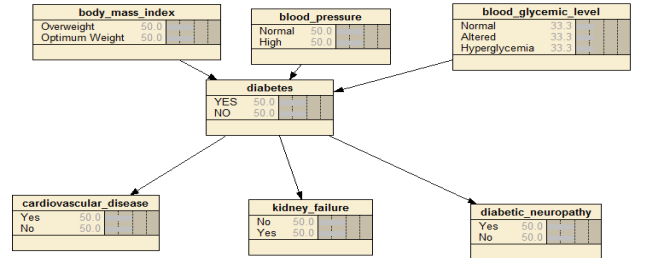


Fig. 1. Diabetes Bayesian network

The composite probability distribution for the presented Bayesian network can be identified as well:

$$P(bmi, bp, bgl, d, cd, kf, dn) = P(bmi)P(bp)P(bgl)P(d|bmi, bp, bgl)P(cd|d)P(kf|d)P(dn|d) \quad (2)$$

Where  $P(bmi), P(bp), P(bgl)$  are marginal probabilities and  $P(d|bmi, bp, bgl), P(cd|d), P(kf|d), P(dn|d)$  are conditional probabilities.

#### IV. RESULTS

Our Bayesian network is built from the qualitative component - structured according to the experts and from the quantitative component of the relationships between the modeled variables. On their basis, we interpreted the inference results on the Bayesian network using Netica simulation environment, by analyzing several cases and observed how the probability variation changes the confidence regarding the most important nodes involved.

For the assigned marginal probabilities above - for parents nodes, for conditional probability distributions for node “Diabetes” (Figure 2) and the main consequences of diabetes based on statistical data:

- ✓ 50% of people with diabetes die primarily of heart disease and stroke;
- ✓ 10-20% of people with diabetes die of kidney failure;
- ✓ damage to the nerves affects up to 50% of people with diabetes;

the compiled Bayesian network looks like in Figure 3.

| body_mass_index | blood_pressure | blood_glycemic_level | YES | NO |
|-----------------|----------------|----------------------|-----|----|
| Overweight      | Normal         | Normal               | 50  | 50 |
| Overweight      | Normal         | Altered              | 50  | 50 |
| Overweight      | Normal         | Hyperglycemia        | 50  | 50 |
| Overweight      | High           | Normal               | 50  | 50 |
| Overweight      | High           | Altered              | 50  | 50 |
| Overweight      | High           | Hyperglycemia        | 50  | 50 |
| Optimum Weight  | Normal         | Normal               | 50  | 50 |
| Optimum Weight  | Normal         | Altered              | 50  | 50 |
| Optimum Weight  | Normal         | Hyperglycemia        | 50  | 50 |
| Optimum Weight  | High           | Normal               | 50  | 50 |
| Optimum Weight  | High           | Altered              | 50  | 50 |
| Optimum Weight  | High           | Hyperglycemia        | 50  | 50 |

Fig.2. Conditional probability distributions of “Diabetes”

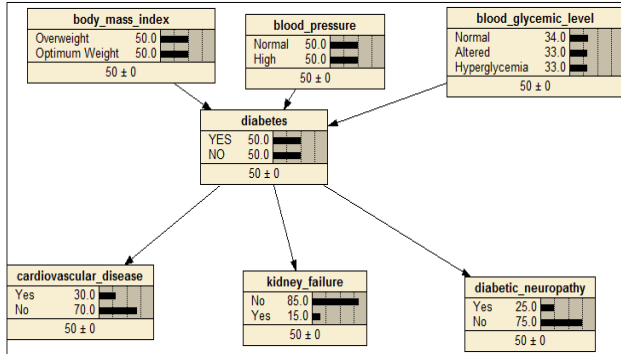


Fig. 3. Compiled Bayesian network

First, we modeled our Bayesian network assigning different values (starting from 60% to 100%) for conditional probabilities having “Blood glycemic level”= *hyperglycemia*. Then, we analyzed our Bayesian network for all the probabilities where variable “Body\_mass\_index”= *overweight*, followed by testing the quantitative component of the network with “Blood pressure”= *high*. All the results are found in Table 1.

TABLE I. PROBABILITY VARIATION CHANGES

| Case | P(d=Yes bmi,bp,bgl)        | P(val) | P(d=Yes)  | P(cd=Yes) | P(kf=Yes) | P(dn=Yes) |
|------|----------------------------|--------|-----------|-----------|-----------|-----------|
| 1.a  | bmi=ov, bp=h, bgl=h        | 60-100 | 50.8-54.1 | 30.3-31.7 | 15.1-15.4 | 25.2-26.2 |
| 1.b  | 1.a + bmi=ov, bp=n, bgl=h  | 60-100 | 51.6-58.3 | 30.7-33.3 | 15.2-15.8 | 25.5-27.5 |
| 1.c  | 1.b + bmi=opt, bp=h, bgl=h | 60-100 | 52.5-62.4 | 31-35     | 15.2-16.2 | 25.7-28.7 |
| 1.d  | 1.c + bmi=opt, bp=n, bgl=h | 60-100 | 53.3-66.5 | 31.3-36.6 | 15.3-16.7 | 26-29.9   |
| 2.a  | bmi=ov, bp=n, bgl=n        | 60-100 | 50.9-54.3 | 30.3-31.7 | 15.1-15.4 | 25.3-26.3 |
| 2.b  | 2.a + bmi=ov, bp=n, bgl=a  | 60-100 | 51.7-58.4 | 30.7-33.3 | 15.2-15.8 | 25.5-27.5 |
| 2.c  | 2.b + bmi=ov, bp=n, bgl=h  | 60-100 | 52.5-62.5 | 31-35     | 15.3-16.2 | 25.8-28.7 |
| 2.d  | 2.c + bmi=ov, bp=h, bgl=n  | 60-100 | 53.4-66.8 | 31.3-36.7 | 15.3-16.7 | 26-30     |
| 2.e  | 2.d + bmi=ov, bp=h, bgl=a  | 60-100 | 54.2-70.9 | 31.7-38.4 | 15.4-17.1 | 26.3-31.3 |
| 2.f  | 2.e + bmi=ov, bp=h, bgl=h  | 60-100 | 55-75     | 32-40     | 15.5-17.5 | 26.5-32.5 |

For a *hyperglycemia* blood level, an *overweight* or *optimum* body mass, and *high* or *normal* blood pressure, a random person is linked to have type 2 diabetes based on probabilities in range [50.8 66.5]% -cases 1.a-1.d.

For an *overweight* body mass index, *high* or *normal* blood pressure, and *normal*, *altered* or *hyperglycemia* blood glycemic level, a random person is associated with an increased risk of diabetes based on probabilities in range [50.9 75]% -cases 2.a-2.f. The results are similar for *high* blood pressure, *overweight* or *optimum* body mass, and *normal*, *altered* or *hyperglycemia* blood glycemic level.

Compared with all the probabilities where “Blood glycemic level”= *hyperglycemia*, cases 2.e and 2.f show the highest values for the effects nodes e.g. as high the probabilities are for these conditional probabilities, as high raise the chances to have diabetes and future illnesses.

“Blood pressure”= *high* is another clue that may indicate a growing in the rate of diabetes and contracting multiple complications. Results showed that increased *overweight* value for the body mass index or increased *high* value for the variable blood pressure is vital in diabetes diagnosis. This conclusion is also valid for increased *hyperglycemia* values but with little smaller probabilities values compared to *overweight* or *high* values.

Another analysis we have made refers to the diabetes criteria consent. The conditional probability in case 1.a actually represents the basic rule for the extended international criteria compliance. The results of modeling this probability can be viewed in Table 1.

If “Blood pressure”= *normal*, for the same variations of probabilities in case 1.a, the results are identical to it and indicate that the presence of blood pressure variable is not a defining criterion like the other two are for the existence or not of diabetes e.g. ADA criteria.

Next, we introduced evidence e.g. information received from external sources, regarding the possible values of a variable, in order to interpret the results of the inference on the Bayesian network. Based on them, the inference results were translated in order to see how these pieces of evidence modify or not the belief on intercept diabetes and how affects the last nodes of our Bayesian network.

For exemple, based on the modeling of the probabilities in case 1.a (starting from 60% to 100%) and for evidences on “Body\_mass\_index” parent node (marginal probability  $P(bmi = overweight) = 100\%$ ), “Blood pressure” parent

node (marginal probability  $P(bp = high) = 100\%$ ) and “Blood glyceic level” parent node (marginal probability  $P(bcl = hyperglycemia) = 100\%$ ), the results for  $P(d)$  are the same as its marginal probabilities (starting from 60% to 100%), for  $P(cd = yes)$  vary from 34 to 50, for  $P(kf = yes)$  from 16 to 20 and for  $P(dn = yes)$  from 28 to 40%.

Furthermore, if “Diabetes” node is certain information, the results for the effects nodes are the same as their marginal probabilities.

The results obtained by adding evidence authenticates the theory related to the rules regarding the transmission of information in a Bayesian network. In other words, for any serial connection, the information is transmitted from the parent node to the child node only if the middle one is uncertain / has no information. In our case, “Diabetes” node has no evidence and  $P(cd|d, bmi, bp, bgl) \neq P(cd|d)$ .

If “Diabetes” node is certain, the information from the parent to the child node is blocked, the probabilities for the children nodes remain the same as their marginals, and  $P(cd|d, bmi, bp, bgl) = P(cd|d)$ .

In all discussed cases, where there was no disease, for all 3 effects nodes, 10% of patients were getting complications and 90% were not getting any of them. Also, we considered  $P(cd = yes|d = yes) = 0.5$ ,  $P(kf = yes|d = yes) = 0.2$ , and  $P(dn = yes|d = yes) = 0.4$ . The rest of the probabilities were equiprobable.

## V. CONCLUSIONS

Causes leading to diabetes are many, varied, and include factors such as physical inactivity, age, high blood pressure, or abnormal cholesterol. Uncontrolled diabetes can reach to skin infections, neuropathy, and even gangrene.

Unlike systems based on certainty factors (production rules), expert systems based on Bayesian networks model uncertainty appealing to probability theory. Bayesian networks give the possibility to model several types of uncertain reasoning, namely causal/predictive, diagnostic/explanatory, or combined.

In this paper, a causal Bayesian network model is used for the analysis of a diabetes case study. The purpose of the present work was to simulate a diabetes Bayesian network starting from three main factors. Based on different probabilities of nodes, we interpreted the compiled Bayesian network in terms of what chances people have to reach diabetes or the effects of it. Our Bayesian network analysis validates the ADA criterion, revealing the tightly bound of body mass index and blood glyceic level, both playing a vital role in the onset of diabetes. Moreover, our study authenticates the theory

related to the rules regarding the transmission of information in a serial Bayesian network.

A potential line of research related to the presented topic can be one of using real probabilities in simulating the Bayesian network to accurately predict the prevalence of diabetes from the cause–effect relationship.

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